

Yizhuo Wang

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Ph.D. Candidate, Multi-Agent Robotic Motion (MARMot) Laboratory, National University of Singapore.
Research interest: robot learning, path planning, reinforcement learning, vision-and-language navigation.

Education

- **Ph.D., Robotics, National University of Singapore** **Singapore**
Advised by Asst. Prof. Guillaume Sartoretti *Fall 2022 – Fall 2026*
 - Research topic: Machine learning based methods for scalable, long-term planning in large multi-agent teams.
- **Master, Mechanical Engineering (Control), National University of Singapore** **Singapore**
Advised by Asst. Prof. Guillaume Sartoretti, GPA: 4.5/5.0 *2020 – 2021*
 - Thesis title: Reinforcement Learning for Multi-Agent Search and Rescue.
- **Bachelor, Mechatronics, Harbin Institute of Technology** **Harbin & Sūzhou, China**
Advised by Assoc. Prof. Ong Chong Jin *2016 – 2020*
 - Thesis title: Development of mini-AGV for cooperative control.

Work Experience

- **National University of Singapore** **Singapore**
Research Engineer *2021 – 2022*
 - Investigate advanced multi-agent planning/exploration methods, funded by Temasek Laboratories @ NUS.
- **Meet Hive Intelligence Co., Ltd.** **Xi'an, China**
Mechanical Development Intern *Summer 2020*
 - Designed the mechanical structure and visual detection system for Garbage Automatic Recycling System for Household (GARSH). Six patents attained.

Project Experience

- **Autonomous Robotic Path Planning**
Reinforcement Learning-based Exploration and Navigation for Mobile Robots *2024 – 2025*
 - Proposed CogniPlan, which employs a conditional generative model for layout prediction and estimates uncertainty to guide downstream path planning. Achieved significant improvements over heuristic/optimization-based and other learning-based SoTA planners; the work was accepted by the top robotics learning conference CoRL.
 - Leveraged expert trajectories to provide soft-constraint rewards and combined them with a hierarchical decision-making framework to enable large-scale autonomous exploration (submitted to T-RO, IROS'25 Workshop).
 - Collaboratively developed a multi-UAV cooperative exploration method under limited field of view, jointly optimizing waypoint selection and UAV orientation (ICRA'25).
 - Collaboratively developed an attention-based autonomous exploration method (ICRA'23, RA-L).
- **Persistent Monitoring for Multi-Objective Optimization**
Spatio-temporal Attention Mechanisms and Collaborative Monitoring Networks *2022 – 2024*
 - Proposed a spatiotemporal attention network for persistent monitoring/reconstruction of multiple moving targets, achieving efficient and balanced long-term performance; the work was published in IROS'23.
 - Extended the method to multi-robot scenarios and achieved joint optimization of path planning and gaze/attitude control (MRS'25 Best Paper, ICRA'23 Workshop).

Multi-Agent Collaboration in Pursuit–Evasion Games

- *Reinforcement Learning-based Visibility-Based Pursuit–Evasion Framework* 2023 – 2024
 - Proposed ViPER, which learns pursuit–evasion strategies under adversarial/worst-case scenarios through visibility modeling and multi-agent reinforcement learning (e.g., privileged learning, attentive critic); the work was published in CoRL'24.
- **Other Directions**
- *Multi-Agent Communication Learning; Physics-based Robotic Affordance Prediction* 2022 – 2023
 - Investigated how multi-agent systems can efficiently exchange and utilize information, and enhanced long-term collaboration and decision-making via a communication memory mechanism (published in AAMAS Journal).
 - Extended a physics-based hierarchical imagination mechanism to infer object categories.
- **Ongoing Research Directions**
- *Vision-Language Navigation for Indoor Scenes; Vision-based UAV Navigation* 2025 – Now
 - Ongoing development for object/text-goal vision-language navigation based on visual semantic segmentation (e.g., SAM 3 / Grounding DINO), combined with language-guided reasoning and commonsense knowledge prediction.
 - Conducting parallel reinforcement learning for UAV navigation using depth cameras within Isaac Lab.

Publications

Conference / Journal / Workshop

- Y. Cao, **Y. Wang**, J. Liang, Y. Zhang, G. Sartoretti. HEADER: Hierarchical Robot Exploration Attention Deep Networks with Expert-Guided Reinforcement Learning. **Submitted to IEEE Transactions on Robotics.** T-RO 2026
- R. Zhao, X. Zhang, Y. Cao, **Y. Wang**, and G. Sartoretti. Attention-based Learning for 3D Informative Path Planning. **Submitted to IEEE International Conference on Robotics and Automation**, Vienna, Austria. ICRA 2026
- X. Zhang, **Y. Wang**, G. Sartoretti. COMPASS: Cooperative Multi-Agent Persistent Monitoring using Spatio-Temporal Attention Network. In *IEEE International Symposium on Multi-Robot & Multi-Agent Systems*, Singapore (Best Paper Award Finalist). MRS 2025
- Y. Cao, **Y. Wang**, J. Liang, G. Sartoretti. Scalable and Expert-guided Reinforcement Learning-based Autonomous Robot Exploration. In *Active Perception Workshop (Best Paper Award)*. IROS 2025
- **Y. Wang**, H. He, J. Liang, Y. Cao, R. Chakraborty, and G. Sartoretti. CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction. In *Conference on Robot Learning*, Seoul, Korea. [Project Page] CoRL 2025
- J. Chiun, S. Zhang, **Y. Wang**, Y. Cao, G. Sartoretti. MARVEL: Multi-Agent Reinforcement Learning for constrained field-of-View multi-robot Exploration in Large-scale environments. In *IEEE International Conference on Robotics and Automation*, Atlanta, USA. ICRA 2025
- **Y. Wang**, Y. Cao, J. Chiun, S. Koley, M. Pham, G. Sartoretti. ViPER: Visibility-based Pursuit-Evasion via Reinforcement Learning. In *Conference on Robot Learning*, Munich, Germany. CoRL 2024
- Y. Cao, R. Zhao, **Y. Wang**, B. Xiang, G. Sartoretti. Deep Reinforcement Learning-based Large-scale Robot Exploration. In *IEEE Robotics and Automation Letters* (vol.9, no.5, pp.4631-4638). RA-L 2024
- **Y. Wang**, Y. Wang, Y. Cao, G. Sartoretti. Spatio-Temporal Attention Network for Persistent Monitoring of Multiple Mobile Targets. In *IEEE/RSJ International Conference on Intelligent Robots and Systems* (pp.3903-3910), Detroit, USA. IROS 2023
- Y. Wang, **Y. Wang**, G. Sartoretti. Full Communication Memory Networks for Team-Level Cooperation Learning. In Springer's *Autonomous Agents and Multi-Agent Systems*, 37, 33. J-AAMAS 2023
- **Y. Wang**, G. Sartoretti. Learning Simultaneous Motion Planning and Active Gaze Control for Persistent

Monitoring of Dynamic Targets. In *Workshop on Active Methods in Autonomous Navigation*. **ICRA 2023**

- Y. Cao, T. Hou, **Y. Wang**, X. Yi, and G. Sartoretti. ARIADNE: A Reinforcement learning approach using Attention-based Deep Networks for Exploration. In *IEEE International Conference on Robotics and Automation* (pp.10219-10225), London, UK. **ICRA 2023**
- Y. Cao, **Y. Wang**, A. Vashisth, H. Fan, and G. Sartoretti. CATNIPP: Context-Aware Attention-based Network for Informative Path Planning. In *Conference on Robot Learning* (pp.1928-1937), Auckland, New Zealand. PMLR. **CoRL 2022**

Selected Patent.....

- Z. Luo, **Y. Wang**, Y. Fu, J. Lu, and H. Zhang, A Rotating Structure for Garbage Sorting. *Utility Model Patent (Chinese)*, CN213474287U. **2021**
- Z. Luo, **Y. Wang**, Y. Fu, J. Lu, and H. Zhang, A Garbage Pushing and Delivery Structure. *Utility Model Patent (Chinese)*, CN213325514U. **2021**
- **Y. Wang**, Y. Fu, Z. Luo, H. Zhang, and J. Lu. Intelligent Recycling Bin (Dys-1). *Design Patent (Chinese)*, CN306247881S. **2020**

Award / Competition

- **Best Paper Award Finalist** **Singapore**
IEEE International Symposium on Multi-Robot and Multi-Agent Systems **2025**
- **Best Paper Award** **Hangzhou, China**
Active Perception Workshop @ IROS2025 **2025**
- **Judge Commendation Award & Championship Award – 5th place** **Singapore**
Singapore Amazing Flying Machine Competition 2023 **2023**
- **National Gold Prize (Red Tour Track)** **Guangzhou, China**
6th China Intl. College Students 'Internet+' Innov. & Entrep. Competition **2020**

Profession Activities

Teaching.....

- **Teaching Assistant** **Singapore**
NUS ME3103 Mechanical Systems Design **Spring 2021**

Paper Review.....

- **Journal Reviewer:** RA-L (2023–), T-ASE (2023–)
- **Conference Reviewer:** AAMAS (2026–), ICRA (2025–), RSS (2024–), CoRL (2023–), IROS (2023–), AAAI (2023–), ECAI (2023–), ISER (2023–), IJCAI (2022–)

Supervised Students.....

- Haodong He (2024–2026), Exchange Student
- Xingjian Zhang (2024–2025), Master of Science → PhD in National University of Singapore.
- Subhadeep Koley (2023–2024), Intern Student → PhD in Lehigh University, USA.
- Xian Yi, (2022–2023), Master of Science → Xiaomi Corp., China.
- Apoorva Vashisth, (2021–2023), Intern Student → PhD in Purdue University, USA.